NETWORKING · NET-05

Cisco IOS CLI Essentials

Use this when: you sit at a Cisco router or switch prompt and need to move between modes, find a command, secure the box and save your work.

BreakFixLearn

CCNA 200-301 · IOS and IOS XE
Sheet v1.0 · Oct 2026

Scan to visit

Command modes: read the prompt

Each mode only accepts its own commands

Prompt	Mode	Notes
`R1>`	**User EXEC**	Basic show, ping
→ `enable`		
`R1#`	**Privileged EXEC**	All show, copy, reload
→ `conf t`		
`R1(config)#`	**Global config**	Settings for the box
→ `int g0/1`		
`R1(config-if)#`	**Interface**	One port; lines use `line vty 0 15`

exit = up one level **end** or **Ctrl+Z** = straight back to `#` **disable** = back to `>`

Help and shortcuts

`?` · `sh?` · `show ?`	What can I type here?
`Tab`	Complete the word
`Ctrl+Shift+6`	Stop a ping or trace
`do show ip int brief`	Run show from config
`no shutdown`	`no` undoes a command

Look before you touch

`show running-config`	Live config (RAM)
`show version`	IOS, uptime, model
`show ip interface brief`	IP and up/up
`show run | section line vty`	One section
`show run | include hostname`	Matching lines

Secure baseline

Replace <secret> · keep owner, model and IOS out of banners

```
hostname R1
no ip domain-lookup   ! typos don't hang
enable algorithm-type scrypt secret <secret>
username admin privilege 15
  algorithm-type scrypt secret <secret>
ip domain-name lab.example.com
crypto key generate rsa modulus 2048
```

```
ip ssh version 2
line vty 0 15
 transport input ssh   ! no Telnet
 login local
 exec-timeout 10 0
banner login # Authorized access only #
! username line is one command
```

Read interface status

show ip int brief

STATUS / PROTOCOL	MEANS
`up / up`	Working
`administratively down`	`shutdown` is set
`down / down`	Layer 1: cable, far end off
`up / down`	Layer 2: encapsulation, clocking

`show interfaces g0/1` adds errors, duplex and speed.

Save and recover

`copy running-config startup-config`
Save. `write memory` does the same.

`reload in 10`
Reboots to the saved config unless you `reload cancel`.

DANGER `erase startup-config` + `reload` wipes it.

CAUTION Type 7 (`service password-encryption`) is easy to reverse.

REMEMBER

- Running = RAM; **save** or lose it
- `secret` beats `password`
- Abbreviate: `sh ip int br`

Sources: Cisco Guide to Harden Cisco IOS Devices · Cisco IOS XE configuration guides · CCNA 200-301 exam topics
CC BY-NC 4.0 · NET-05 v1.0